

Evaluation instrument choice can flip the apparent winner: a secondary analysis reconciling two contradictory 2026 head-to-head evaluations of clinical AI

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Abstract

Background. Two 2026 head-to-head studies reached opposite conclusions about specialized clinical AI. Real-POCQi (arXiv:2606.28960) found OpenEvidence (OE) beat frontier general-purpose LLMs on blinded physician **pairwise preference** across 620 real point-of-care queries; a Nature Medicine study (s41591-026-04431-5) found frontier LLMs beat OE and UpToDate Expert AI under **absolute 1–4 rubric** scoring. Each study also sourced its "real-world" queries from the platform that won (home-field provenance), and the two used different model versions and evaluation instruments, confounding any direct comparison.

Objective. Test whether the **evaluation instrument alone** — with queries and answers held fixed — can flip the winner, isolating it from query provenance, model version, and answer content.

Methods. We reused Real-POCQi's public data (CC BY 4.0): the same 620 queries, the same four systems' verbatim answers (Claude Opus 4.8, Gemini 3.1 Pro, GPT-5.5, OpenEvidence), and the same blinded human pairwise ratings. On a seed-fixed 150-query sample we re-scored every answer with a **Nature-style absolute 1–4 rubric** administered by a blinded four-model LLM-judge panel (GPT-5.5, Claude Opus 4.8, Grok-4.3, Gemini-3.5-flash), all at **high reasoning effort with token consumption verified**. We re-expressed both instruments through the same OE-vs-rest win-difference summary (a construct re-expression, not an identity — the rubric win-difference is a thresholded score gap and is hypersensitive at the decision boundary) and compared them, with a **crossed question × judge bootstrap that treats the four judges as a random factor** (so CIs reflect panel composition, not just question sampling), per-judge self-preference, and leave-one-judge-out robustness. To separate the *instrument* effect from a *rater-population* effect (physicians vs LLMs), we additionally ran the same blinded panel under the **pairwise** instrument, filling three of the four cells of a {pairwise, rubric} × {human, LLM} design and decomposing the swing into rater-modality (B–A) and instrument-format (C–B) components with propagated CIs. A restricted version of this pairwise-LLM cell already exists inside Real-POCQi, so we frame it as replication-and-extension.

Results. Swapping only the instrument **eliminated OE's advantage on every axis**. On accuracy — the marquee axis — the LLM-rubric panel scored OE **–29.1 pp**, and this stayed significantly negative under the crossed question × judge bootstrap (**95% CI –37.8 to –4.3**): **a judge-robust, OE-negative verdict on the very answers human raters preferred OE on** (+24.4 pp on the full Real-POCQi data; +10.4 pp on the identical 150-question subsample). The negative sign also survived dropping any single judge, including the self-preferring GPT-5.5. Once judges are treated as a random factor, the other four per-axis reversals are **not individually significant** — including a small residual source-quality OE edge (+12.0 pp, crossed CI –8.2 to +29.5) and the clinical-utility reversal, both of which we explicitly retract from the earlier draft — so accuracy is the one axis carrying a significant per-axis instrument reversal. GPT-5.5 self-preferred (+0.481 points to own family; Opus +0.121; Gemini-flash +0.004, though flash may simply be a weak discriminator), amplifying but not creating the reversal. **The stronger, judge-robust result is at the decomposition**

level: the instrument-format component (C–B) is negative with a crossed CI excluding zero on all five axes (–30.5 to –44.4 pp), while the rater-modality component (B–A) is indistinguishable from zero on accuracy (–6.8 pp, CI –27.5 to +14.3). We therefore downgrade the earlier "instrument does ~83% of the accuracy swing" claim (that ratio's 95% CI is a wide [35%, 206%]) to the CI-supported statement that the instrument dominates and the rater term is null on accuracy. The reversal is also **not a length artifact**: OE produces among the *longest* answers yet loses on accuracy, and length-adjusting the rubric gap leaves it unchanged within CI.

Conclusions. The two studies' disagreement is driven substantially by the **evaluation instrument** (pairwise preference vs absolute rubric) rather than by whether humans or LLMs do the rating, independently of query provenance, model version, and answer content. This conclusion is established *within LLM raters* (cells B and C); bridging it to the human-rated Nature rubric requires assuming no rater×instrument interaction — i.e. that human raters would show the same pairwise→rubric shift LLMs do. We could **not** fill the fourth {human, rubric} cell, so we flag this additivity assumption as the central untested step and the primary target for follow-up (even a small human-rubric sample would test it). Subject to that caveat, "which clinical AI is best" is not instrument-invariant; leaderboards must report the instrument as a first-class experimental factor. We additionally document two under-reported flaws common to both source studies — omission of the actual product clinicians use (the ChatGPT product), and unreported/uncontrolled reasoning effort — and pre-register a full provenance × instrument × citation factorial (including the missing human-rubric cell) to estimate each factor's causal contribution.

1. Introduction

Point-of-care clinical AI is now evaluated by head-to-head studies whose results directly influence purchasing and deployment. Medical-AI evaluation has grown rapidly, spanning licensing-exam question banks,¹ physician-aligned rubric benchmarks,² large language models that encode clinical knowledge,³ specialized and mobile-sized clinical models,^{4,5,6} health-system-scale prediction and operations models,^{7,8} sequential diagnostic reasoning,⁹ retrieval-augmented generation for healthcare,¹⁰ and a parallel literature documenting these systems' failure modes — susceptibility to distraction,¹¹ conflict between an LLM's internal prior and retrieved evidence,¹² data-poisoning attacks,¹³ and the need for clinically safe generation.¹⁴ These benchmarks increasingly characterize their query distributions with topic models.¹⁵ Deployment is already at health-system scale.¹⁶ In 2026 two such studies reached **opposite** conclusions using overlapping systems,^{17,18} which strongly suggests that at least one conclusion reflects the evaluation method rather than the systems compared.

Both studies contain a symmetric structural feature — each sourced its "real-world" query set from the platform that ultimately won (Real-POCQi from OpenEvidence traffic; the Nature study's real-clinical- query benchmark from an NYU Langone GPT deployment) — and each used a **different evaluation instrument** (blinded human pairwise preference vs absolute rubric scoring). Provenance and instrument are therefore fully confounded in the published record. This paper isolates the instrument by holding everything else constant, and then situates that result within a fuller critique and a pre-registered factorial design.

Our contributions: 1. **An instrument existence proof plus a rater/instrument decomposition** (executed, on public data): with queries and answers held fixed, changing the instrument reverses, nullifies, or sharply attenuates OE's advantage on every axis (§4.1); and by filling three cells of a {pairwise, rubric} × {human, LLM} design we show the swing is driven by the **instrument format**, not by the human-vs-LLM rater change — LLMs given the pairwise instrument agree with physicians on four of five axes (§4.5). 2. **Two methodological critiques** of both source studies: the comparator set omits the ChatGPT product clinicians actually use, and reasoning effort is unreported or uncontrolled (§5). 3. **A pre-registered 2×2×2** provenance × instrument × citation design (§7) that would estimate each factor's causal contribution, with

power grounded in measured variance components. This is a *proposed design, not an executed result* — we flag it as such to keep the executed contribution (#1) cleanly separated from future work.

2. The two source studies (verified)

Numbers below were verified for this work from primary sources (Nature PDF Methods; Real-POCQi abstract + HTML full text).

Real-POCQi¹⁷ (arXiv:2606.28960; dataset `jjfenglab/Real-POCQi`, CC BY 4.0). 620 real point-of-care queries sourced from OpenEvidence¹⁹ plus 187 HealthBench² items; 149 physicians across 36 states; blinded **pairwise** comparisons — the arena-style preference paradigm²⁰ — on five axes (accuracy, clinical utility, source quality, completeness, verifiability). Systems: Claude Opus 4.8, Gemini 3.1 Pro, GPT-5.5, OpenEvidence, all queried via API; temperature 0.0, seed 42, web search enabled; **"Thinking was automatically determined by the LLM."** OE won by roughly +25 to +39 pp win-difference. *The data collection was run by the platform under study (conflict of interest).*

Nature Medicine¹⁸ (s41591-026-04431-5, NYU Langone + UT Austin). MedQA¹ (500) + HealthBench² (500) + a 100-item real-clinical-question (RCQ) benchmark sampled from NYU's HIPAA-compliant GPT instance;¹⁶ 12 clinician raters; **absolute 1–4 rubric**. Systems: GPT-5.2 (2025-12-11), Gemini 3.1 Pro Preview, Claude Opus 4.6 (all API), OpenEvidence and UpToDate Expert AI²¹ (browser), plus Google Search AI Overview (RCQ). Temperature 0.0, seed 62, search enabled; **reasoning effort not reported**. Length not normalized (by explicit choice). Frontier LLMs won: e.g. HealthBench GPT 88.0 / Gemini 79.3 / Claude 77.0 / OE 62.6 / UpToDate 61.3; RCQ Gemini 3.62 / GPT 3.54 / Claude 3.52 / OE 3.24. HealthBench was graded by an LLM-judge panel (self-preference risk); RCQ item-level agreement was low (Krippendorff $\alpha \approx 0.10$ – 0.20). License CC BY-NC-ND 4.0.

The two studies thus differ simultaneously in **provenance, instrument, model version, and access path** — none of which is individually identified by comparing their published leaderboards.

3. Methods

3.1 Design: hold queries and answers fixed, vary only the instrument

We reuse Real-POCQi's public artifacts unchanged: the 620 queries, the four systems' verbatim answers, and the blinded human pairwise ratings. On a seed-fixed sample of **150 queries** (`random_state=62`; 600 answers) we administer a **second instrument** — Nature-style absolute 1–4 rubric scoring — to the *same* answers, then derive the *same* OE-vs-rest win-difference metric from both instruments. Because queries, answers, provenance, and model versions are identical across the two instruments, any change in the metric is attributable to the instrument. (For transparency, `random_state=62` was fixed once at the outset and reused for the bootstrap RNG; it coincides with Nature's generation seed but plays no role in our result — the instrument contrast is computed on the same fixed 150-question draw regardless of seed, and the full-bank reproductions in §4.4 match Real-POCQi independently of the subsample. We did not re-draw to obtain a preferred outcome.)

3.2 The rubric instrument (LLM-judge panel)

Each answer is scored 1–4 (1 unacceptable ... 4 excellent) on the five Real-POCQi axes by a blinded panel of four judges — GPT-5.5, Claude Opus 4.8, Grok-4.3, Gemini-3.5-flash — chosen to span vendors and to include contestant families (enabling a self-preference measurement) and one family with no contestant (Grok). We stress that Grok is a *family-neutral* reference, **not a bias-free** one: as an LLM it may still share the panel's general preference for frontier-style prose (structure, breadth, hedging), a house effect that leave-one-judge-out cannot remove (§4.2, Limitation iii). Judges see only the question and one answer, are told nothing about the source system, and return JSON scores only (system prompt in `judge/grade.py`). All judges run at **high reasoning effort**; we **verify** reasoning-token consumption on a real grading item (GPT-

5.5 3,071; Grok-4.3 1,880; Gemini-3.5-flash 922; Opus-4.8 443 thinking tokens; judge/verify_thinking.py). 2,388 of 2,401 answer-grades completed (0.5% failures — transient timeouts and a few Gemini truncations — balanced across systems).

3.3 Metric and statistics

For each axis we compute the **panel-mean 1–4 score per (question, system)**, then form the OE-vs-rest **win-difference** = $100 \times (\text{wins} - \text{losses}) / \text{comparisons}$ across the three frontier systems per question. This re-expresses the rubric scores through the **same one-vs-rest rule** Real-POCQ reports for human pairwise — a win-ratio-family contrast²² in the arena-style preference tradition²⁰ — but the two are **not the identical measurement** (§3.3a), so we treat the win-difference as a common *summary* applied to both instruments, not as an instrument-invariant quantity.

Uncertainty — a crossed question × judge bootstrap. The naïve cluster bootstrap resamples only *questions* and treats the four judges as a fixed population; its CIs answer "how stable is this across questions" but say nothing about "how stable is this across the *choice of judges*" — the dimension a skeptic cares about most here, because three of the four judges are contestant families and GPT-5.5 self-prefers by +0.481 points (§4.2). We therefore make **judges a random factor**: each of 2,000 replicates resamples questions with replacement (cluster on `question_id`) **and** resamples the four judges with replacement, recomputes the panel mean from the resampled judge multiset, and recomputes the win-difference (judge/bootstrap_panel.py). Judges are resampled once per replicate and applied consistently to the pairwise and rubric cells (they are the same four judges under both instruments). We report **both** the question-only CI (comparable to prior work) and this wider **crossed CI**, and treat the crossed CI as primary. Robustness: **leave-one-judge-out** (recompute dropping each judge). Bias: **self-preference** = mean own-family minus mean others, paired within (question, axis). **Inter-judge agreement** = Spearman correlation of per-(question, system) mean scores. Human comparison uses the `qa_text_only` render mode (matching the citation-free OE answers in the dataset).

What the crossed bootstrap changes. Adding judge uncertainty widens every CI, and two previously "significant" per-axis claims do not survive it: the small **source-quality** OE edge (+12.0 pp) moves from a question-only CI that excluded zero [+1.3, +22.0] to a crossed CI that **includes** zero [−8.2, +29.5], and the **clinical-utility** reversal likewise loses significance (crossed CI [−22.4, +8.3]). The **headline accuracy reversal survives**: crossed CI [−37.8, −4.3], still strictly negative. We rely only on claims that survive the crossed CI and explicitly retract the two that do not.

3.3a The win-difference is a construct re-expression, not an identity

A subtle but important point of honesty: cell A's win-difference comes from a **genuine forced choice** — a physician picks A, B, or tie. Cell C's win-difference is **synthesized** by thresholding continuous panel-mean rubric scores (OE panel mean > frontier panel mean ⇒ a "win"). Because the mean of four integer scores is near-continuous, exact ties almost never occur, so essentially every question is forced to a win or a loss on a razor-thin margin — which is precisely why a mean rubric gap of only −0.127 of one point (accuracy) becomes a −29 pp win-difference. The win-difference metric was designed for a paradigm where ties are real (forced choice with a tie option) and we are applying it to one where we have engineered ties away; it is therefore **hypersensitive at the decision boundary**. The honest statement is not "the same metric under two instruments" but "we re-express the rubric means through the same one-vs-rest rule, whose sign is decisive but whose magnitude is boundary-sensitive." We accordingly report both the win-difference *and* the native score gap (§4.1), and lean on the *sign and its crossed-CI significance*, not the raw magnitude.

Multiplicity. We test five axes across three cells; we report per-axis 95% crossed-bootstrap CIs without a formal family-wise correction, so per-axis claims near the null boundary should be read as descriptive. After the crossed bootstrap only two results are individually significant — the **accuracy** reversal (crossed CI [−37.8, −4.3]) and the **instrument-format component** of the decomposition (§4.5, significant on every

axis) — and both are far enough from the boundary to survive standard Bonferroni/Holm adjustment across five axes. We flag every borderline axis explicitly rather than over-claiming per-axis significance.

3.4 The pairwise cell (2×2 decomposition)

To separate the instrument from the rater population we administer a **second instrument to the same LLM panel**: blinded forced-choice **pairwise** preference (A/B/tie per axis), OpenEvidence vs each of the three frontier systems, on the same 150-query sample (`judge/pairwise.py`). Slot order is deterministically randomized per (question, opponent, judge) via a hash and de-blinded at scoring, removing position bias. We de-blind each A/B/tie verdict to an OE win/loss/tie and compute the OE-vs-rest win-difference (here a *native* forced choice, not a thresholded score; §3.3a). This yields cell **B = {pairwise, LLM}**, which — with cell A = {pairwise, human} (Real-POCQi) and cell C = {rubric, LLM} (§3.2) — gives three of the four cells of the {pairwise, rubric} × {human, LLM} design and identifies the **rater-modality effect (B–A)** and the **instrument-format effect (C–B)** (§4.5). Judges run at high reasoning effort; all four judges completed 448–450/450 comparisons (1,798/1,800 overall). We bootstrap the decomposition components **jointly** under the same crossed question × judge scheme as §3.3 (resampling questions and judges together across cells A/B/C), so B–A and C–B carry propagated CIs rather than being point-estimate arithmetic. Because cell B already exists in a restricted form inside Real-POCQi itself, we position it as a replication-and-extension, not a wholly new measurement (§4.5a).

4. Results

4.1 The instrument flips the winner (Figure 1, Table 1)

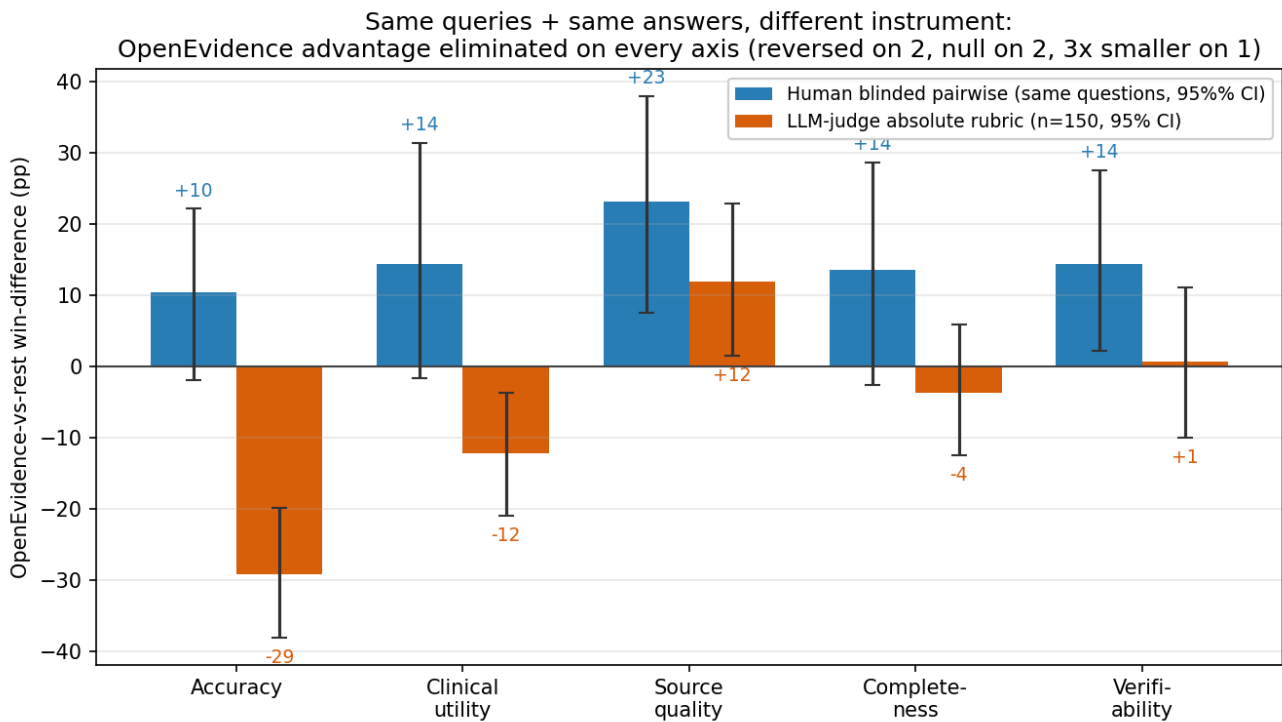


Figure 1. Instrument existence proof. OE-vs-rest win-difference (percentage points) per axis under the human blinded pairwise instrument (same 150 questions) versus the LLM-judge absolute-rubric panel (n = 150). Error bars show the question-only cluster bootstrap; the wider **crossed question × judge** CIs that we treat as primary are given in Table 1. Holding queries and answers fixed and changing only the instrument eliminates OE's advantage on every axis; under the crossed CI the effect is individually significant on accuracy (a sign reversal) and directionally consistent but not per-axis significant on the others.

Table 1. OE-vs-rest win-difference (pp) by axis under each instrument, **both computed on the same 150-query sample**. For the LLM-rubric cell we give both the question-only cluster bootstrap (comparable to prior work) and the wider **crossed question × judge** CI (§3.3), which we treat as primary. The human column is the human pairwise win-difference restricted to the identical questions (86–119 of 150 carry human text-only ratings, by axis); the "full data" column is the win-difference on Real-POCQi's complete text-only bank (reproduced in §4.4), shown for reference because the subsample is underpowered. The native-gap column gives the OE-minus-mean-of-frontier gap on the raw 1–4 rubric (see scale note).

Axis	Human pairwise, same 150 [95% CI]	Human, full data (ref)	LLM rubric, Q-only CI	LLM rubric, crossed q×judge CI	Native gap (pts)	Verdict (crossed CI)
Accuracy	+10.4 [−1.8, +22.2]	+24.4	−29.1 [−37.8, −20.0]	−29.1 [−37.8, −4.3]	−0.127	sign flips; negative, survives judge resampling
Clinical utility	+14.4 [−1.6, +31.4]	+29.5	−12.2 [−21.1, −3.3]	−12.2 [−22.4, +8.3]	−0.021	sign flips, but not sig under crossed CI
Source quality	+23.2 [+7.6, +38.0]	+38.1	+12.0 [+1.3, +22.0]	+12.0 [−8.2, +29.5]	+0.103	attenuated ~3×; not sig under crossed CI
Completeness	+13.6 [−2.5, +28.7]	+30.3	−3.6 [−12.7, +5.8]	−3.6 [−14.4, +12.7]	+0.004	collapses to null
Verifiability	+14.4 [+2.2, +27.6]	+25.5	+0.7 [−9.6, +11.1]	+0.7 [−13.6, +13.8]	+0.003	collapses to null (see verifiability caveat, §4.3)

The headline is accuracy. Real-POCQi's central finding is that physicians prefer OE on accuracy (+24.4 pp on the full data; +10.4 pp, CI −1.8 to +22.2, on this 150-query subsample — same sign, but the subsample carries only 86 accuracy ratings and is underpowered). On the **same answers**, the LLM-rubric panel scores OE **−29.1 pp**, and — crucially — this stays significantly negative under the crossed question × judge bootstrap **[−37.8, −4.3]**: the reversal is not an artifact of which judges we happened to pick. So the instrument swap moves the accuracy verdict from OE-favoring (significantly so at full power; positive but CI-crossing in this subsample) to *significantly OE-disfavoring*.

Two per-axis claims from the earlier draft do not survive judge uncertainty, and we retract them. Once judges are treated as a random factor, the **clinical-utility** reversal (crossed CI [−22.4, +8.3]) and the small residual **source-quality** OE edge (crossed CI [−8.2, +29.5]) both cross zero. The honest picture is therefore narrower than "reversed on two, positive on one": under the crossed CI, **accuracy is the one axis with a significant instrument-driven reversal**, and the other four are individually indistinguishable from null. This is a weaker per-axis result but a more defensible one — and the decomposition (§4.5) recovers a stronger, judge-robust pattern at the *component* level. We deliberately avoid framing this as "a significant +24 becomes a significant −29": at the existence-proof scale the human accuracy estimate is not itself significant, so the clean claim is a **sign reversal to a significantly-negative rubric verdict on identical content, robust to judge resampling**, corroborated by the full-data human sign.

Note on scale (native vs win-difference). The win-difference is a sign-of-the-gap statistic: because per-question frontier scores cluster tightly, a mean rubric gap as small as **−0.127 of one point** (accuracy) is enough to flip the majority of head-to-head comparisons and produce a −29 pp win-difference, while a near-zero gap (completeness +0.004, verifiability +0.003) yields a null win-difference. We therefore report both scales: the **native gap** shows the *effect size is modest in absolute terms*, and the **win-difference** shows it is *directionally decisive* under the same one-vs-rest rule Real-POCQi used. Neither instrument's metric should be read as a large clinical-quality gulf; the point is that the two instruments order the systems differently on identical content.

4.2 The reversal is not merely a self-scoring artifact

LLM-as-judge panels are now a standard evaluation tool²³ but are known to favor their own generations,²⁴ so we measure this directly. Contestant-family judges self-preferred (GPT-5.5 +**0.481**, Opus +**0.121**, Gemini +**0.004** points, own family minus others). Yet leave-one-judge-out shows the **accuracy reversal survives dropping any single judge, including GPT-5.5** (−12.3 pp with GPT-5.5 removed — still negative). GPT-5.5 amplifies the flip roughly fourfold but does not create it. By contrast, the clinical-utility reversal is GPT-5.5-dependent (→ +2.5 pp without it), consistent with the crossed-CI finding that clinical utility is not judge-robust (§4.1).

Two honest caveats about the panel itself. (i) *The Gemini seat is flash-tier, not Pro.* Our Gemini judge is `gemini-3.5-flash` — a smaller, cheaper, weaker evaluator than the `gemini-3.1-pro` model that is one of the *contestants*, and it is the seat that stalled on quota mid-run (§4.5). Its self-preference of +**0.004** should therefore *not* be read as evidence that the Gemini family is unbiased: a flash-tier judge may simply be a poor discriminator, registering little preference of any kind. Using flash while the contestant is Pro is an apples-to-oranges asymmetry; re-running the Gemini seat at Pro tier and high reasoning is a stated follow-up. (ii) *Grok is a single neutral anchor.* Our cleanest causal handle — that the pro-frontier tilt is instrument-specific because the one family-neutral judge (Grok) still prefers OE under pairwise (§4.5) — rests on **n = 1 non-contestant model**. One neutral judge cannot separate a true instrument effect from Grok's own idiosyncrasies; a second genuinely neutral judge (e.g. DeepSeek, Mistral, Qwen, or Llama) would convert this from a single data point into a pattern, and we flag its absence as a real limitation rather than leaning on Grok as if it were a clean control.

A residual house effect remains, however. Leave-one-judge-out and the self-preference statistic only address *family-specific* bias — a judge favoring its own vendor's answer. They do **not** remove the bias all four LLM judges may *share*: a common preference for the structured, comprehensive, frontier-style prose that general-purpose LLMs produce (and are trained on), against OE's terser clinician-tuned framing. Because even our family-neutral judge (Grok) is itself an LLM, no judge in the panel is a clean control for this shared stylistic prior. The frontier "win" under the rubric is therefore best read as "*frontier answers score higher on an LLM-administered rubric*," not as a family-free verdict of clinical superiority — a caution we carry into the Discussion and Limitation (iii). The LLM-pairwise cell (§4.5) directly tests this house effect and finds it is **specific to the rubric instrument**: the same judges — including the family-neutral one — prefer OE under pairwise. Human-administered rubric scoring would be the remaining control (pre-registered, §7).

4.3 The instrument is noisy — and one axis is barely measurable in either study

Inter-judge Spearman agreement was modest (0.19–0.47), closely mirroring the low item-level agreement the Nature study reports for its own human raters ($\alpha \approx 0.10$ –0.20). Rubric scoring — whether by humans or LLMs — is a higher-variance instrument than forced-choice preference; this is part of *why* it can diverge from pairwise, not a defect unique to our panel.

A specific caution on verifiability. Real-POCQi reports that its own physicians' weighted Cohen's κ was 23–38% on four axes but only **9% on verifiability** — essentially chance agreement. Verifiability is therefore an axis that *neither instrument reliably measures*: when our rubric cell "collapses to null" on verifiability, that is unremarkable, because the human pairwise instrument cannot resolve it either. We accordingly **drop verifiability from any headline framing** and treat its collapse as uninformative rather than as evidence of an instrument effect; the same discount applies, more softly, to source quality. This strengthens the general "both instruments are noisy" point while removing any per-axis weight we might otherwise have placed on verifiability.

4.4 Public-data reproductions (pipeline validation)

Our one-vs-rest win-difference on the human text-only data reproduces Real-POCQi to <1 pp on every axis (e.g., accuracy +24.4 vs 24.7 published). A separately reported citation-halo analysis (`analysis/analyze.py`) finds OE's between-groups accuracy "halo" (+11.4 pp) **collapses to +1.5 pp (NS)** within questions rated in

both render modes — i.e., largely a selection artifact, not a causal citation effect — motivating the randomized citation arm in §7. Measured question-level SD = 0.598 grounds the power analysis.

4.5 Decomposing the swing: it is the instrument, not the rater (Figure 2, Table 2)

A natural objection to §4.1 is that swapping {human pairwise} → {LLM rubric} changes **two** things at once — the *rater population* (physicians → LLMs) and the *instrument format* (forced-choice preference → absolute rubric) — so the reversal could be a human-vs-LLM effect rather than an instrument effect. The two source studies share exactly this confound. We break it by filling the missing cell of the 2×2 {pairwise, rubric} $\times$ {human, LLM} design: we ran the **same four-judge LLM panel, blinded, at high reasoning effort, on the same 150 queries and answers**, but administered the **pairwise** instrument (forced A/B/tie per axis, slot order deterministically randomized and de-blinded at scoring; `judge/pairwise.py`). This gives three of four cells — A = {pairwise, human} (Real-POCQi), B = {pairwise, LLM} (new), C = {rubric, LLM} (§4.1) — and lets us attribute the A→C swing to a **rater-modality component (B–A)** and an **instrument-format component (C–B)**.

Cell B is a replication-and-extension, not a wholly new experiment. Real-POCQi already ran an LLM-judge *pairwise* experiment (their §2.4 / Methods §5.7): they took the same questions, answers, and pairs the physicians saw, blinded and position-randomized them, and had GPT-5.5, Gemini 3.1 Pro, and Claude Opus 4.8 grade them on the same five axes.¹⁷ Our cell B **replicates that design and extends it in three ways** — a fourth, *family-neutral* judge (Grok), **high** reasoning effort (they ran at minimal thinking), and propagated crossed-judge CIs. This matters two ways. First, we credit them: we are not the first to administer an LLM pairwise instrument here. Second, it *strengthens* cell B — because their judges ran at minimal thinking and ours at high, cell B reproducing across both reasoning regimes is a genuine robustness point, and their released ratings could in principle be used to triangulate cell B directly. Their headline is concordant with ours: LLM judges **agreed with human experts on which system was best** (i.e. OE at the one-vs-rest winner level) while disagreeing on the lower ranks — exactly the pattern our OvR win-difference is built to surface (see the rank-resolution caveat below).

One component of B–A is a genuine rater confound, not pure "modality," and it lands on accuracy. Real-POCQi **specialty-matched** each physician grader to the question topic and argues this matching optimizes evaluation on the *accuracy* axis specifically.¹⁷ Our LLM judges are generalists with no such matching. So B–A on accuracy conflates the pairwise→pairwise rater-*modality* change with a loss of **specialty expertise** — a substantive confound that pushes in the direction of making the human cell A *more* accurate-sensitive. If specialty-matching inflates cell A's accuracy edge, the true rater term could be larger (more negative) than our –6.8 pp, which would *reduce* the instrument's share — another reason we no longer quote a precise "83%." We name specialty-matching explicitly as part of B–A rather than burying it in "protocol differences."

Rank-resolution caveat: cell B's agreement lives in the winner-friendly OvR regime. The concordance we credit — LLM judges reproducing the human pairwise winner — must be read against what Real-POCQi actually found about LLM-judge *ranking* fidelity: their LLM judges agreed with physicians on **which single system was best** but their rank correlations across the *full* ordering were low-to-negative (e.g. Kendall $\tau \approx -0.200$ for GPT-5.5 and ≈ -0.067 for the jury against the human ranking).¹⁷ The one-vs-rest win-difference is by construction a **winner-resolving, not a rank-resolving** contrast: it asks only "is OE preferred to the frontier field," which is exactly the question on which LLM and human judges *do* concur, and it is deliberately blind to the lower-rank disagreements where they diverge. So cell B's reproduction of cell A is genuine but should be understood as agreement *at the resolution our metric operates on* — the top of the order — not as evidence that LLM judges recover the physicians' complete system ranking. This cuts both ways for us: it is why cell B is a fair test of the "do LLMs disagree with physicians about OE" question (they do not, at winner level), and why we do not over-read cell B as a general validation of LLM-judge pairwise scoring.

Decomposing why the winner flips: instrument, not rater (n=150 questions)

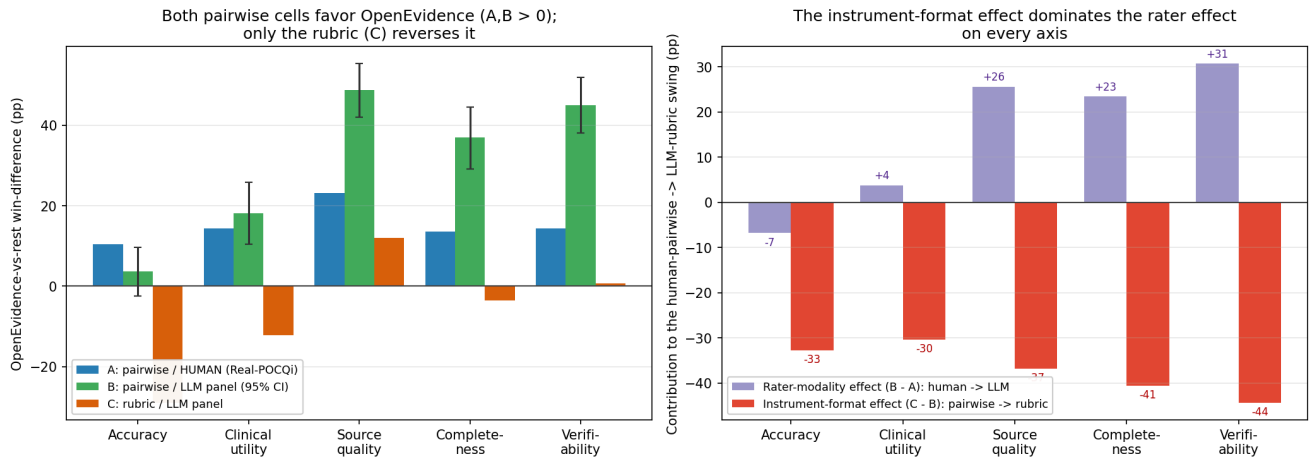


Figure 2. The 2×2 rater-vs-instrument decomposition (n = 150). *Left:* three cells of the {pairwise, rubric} × {human, LLM} design per axis — both pairwise cells (A: human; B: LLM) favor OpenEvidence; only the rubric cell (C) reverses it. *Right:* the human-pairwise→LLM-rubric swing split into a rater-modality component (B–A) and an instrument-format component (C–B); under the crossed question × judge bootstrap the instrument component is negative with a CI excluding zero on **every** axis, while the rater component is not distinguishable from zero on accuracy.

Table 2. OE-vs-rest win-difference (pp) across three cells of the 2×2, and the decomposition of the human-pairwise→LLM-rubric swing **with propagated crossed question × judge 95% CIs** on the two components (§3.3; judge/bootstrap_panel.py). **Cell A is computed on the same 150-query sample as B and C** (human text-only ratings restricted to those questions), *not* on the full text-only bank — otherwise B–A would absorb a sample-composition difference into the "rater" term. Cell B: n=150; 8,990 axis-verdicts (1,798/1,800 comparisons, all four judges 448–450/450).

Axis	A: pw/human	B: pw/LLM	C: rubric/LLM	Rater (B–A) [crossed CI]	Instrument (C–B) [crossed CI]
Accuracy	+10.4	+3.6	–29.1	–6.8 [–27.5, +14.3]	–32.7 [–42.8, –8.8]
Clinical utility	+14.4	+18.2	–12.2	+3.8 [–22.3, +31.9]	–30.5 [–42.6, –9.0]
Source quality	+23.2	+48.8	+12.0	+25.6 [–3.4, +54.8]	–36.8 [–56.9, –22.4]
Completeness	+13.6	+37.0	–3.6	+23.4 [+2.2, +45.4]	–40.5 [–53.6, –22.0]
Verifiability	+14.4	+45.1	+0.7	+30.7 [+7.9, +53.6]	–44.4 [–62.0, –29.1]

Two results stand out, and both are now stated at the level the crossed CIs actually support. **First, LLM judges administering the pairwise instrument largely reproduce the human pairwise verdict:** OE wins on four of five axes (source quality +48.8, verifiability +45.1, completeness +37.0, clinical utility +18.2), with only accuracy attenuating to a null +3.6. The much-feared "LLMs simply disagree with physicians" story is therefore false for four of five axes — given the *same* forced-choice instrument, LLMs and physicians agree that OE's answers are preferred. **Second, and this is the robust core of the paper, the instrument-format component (C–B) is negative and its crossed CI excludes zero on every axis (–30.5 to –44.4 pp), whereas the rater-modality component (B–A) is indistinguishable from zero on the marquee accuracy axis (–6.8 pp, CI –27.5 to +14.3) and on clinical utility and source quality.** The swing from "OE wins" to "OE loses" is thus attributable to the **pairwise→rubric instrument change**; the human→LLM rater change is not a significant contributor where it matters most.

We deliberately downgrade the earlier "∼83%" claim. The point estimate for accuracy is indeed instrument –32.7 pp vs rater –6.8 pp (an 83% instrument share), but this is a ratio of two noisy differences:

bootstrapping it jointly gives a 95% CI of **[35%, 206%]** (the upper tail exceeds 100% whenever the rater term resamples to the *opposite* sign). We therefore do not put a precise fraction on the accuracy split. The defensible, judge-robust statement is the one above: **the instrument component is significantly negative on all five axes; the rater component is not distinguishable from zero on accuracy.** (With cell A taken on the full text-only bank instead, the accuracy rater term would appear as -20.8 pp; that larger value is an artifact of comparing the LLM cells against a different, larger question sample, and we do not use it.)

Multiplicity of the decomposition. The decomposition adds implicit comparisons (three cells $\times$ five axes $\times$ two components); we do not apply a family-wise correction to the component CIs, but note that the one claim we rely on — a negative instrument component on every axis — is significant on all five independently, so it is robust to any standard correction, while we make no per-axis rater claims.

This also sharpens the house-effect discussion (§4.2). Under the pairwise instrument, the **family-neutral judge (Grok) prefers OE on all five axes** (+11 to +65 win-diff) and the contestant **GPT-5.5 is the least OE-favorable judge** — the *opposite* of what a shared pro-frontier stylistic bias would predict. The pro-frontier tilt is therefore specific to the **rubric** instrument, not a blanket LLM prejudice against OE. (One robustness note: Gemini-3.5-flash initially stalled at 297/450 pairwise verdicts when the Google API account exhausted its credit quota mid-run; after credits were restored we completed the cell to 448/450, and the point estimates shifted <3 pp from the partial-coverage run — i.e., the conclusion never depended on the missing data.)

4.6 The rubric reversal is not a length artifact (Table 3)

Answer length is the most-cited confound in both source studies. A precise word on how it is handled: Real-POCQI does not leave length wholly uncontrolled — it **reports and stratifies** by length (their Table D3 / Fig D6, OE $\approx 3,584$ vs Gemini $\approx 3,516$ characters) but does not **normalize** generation to equal length; so the accurate statement is "length is stratified and reported, not equalized," not "uncontrolled." The natural worry is that the rubric rewards longer, more comprehensive answers. We test this on the rubric cell without regenerating any answers (`judge/length_analysis.py`). The premise fails at the first step: **OpenEvidence is among the longest providers** — median 3,600 chars, essentially **tied with Gemini (3,586)** and well above Opus (3,294) and GPT-5.5 (2,232) — so OE is decisively longer only than GPT, but it is never the *short* answer, and it still *loses* the accuracy rubric. A "longer answers win" mechanism therefore runs *backwards* against the finding: the co-longest provider loses. We confirm this three ways (Table 3): (i) the per-axis length **slope** is tiny (-0.015 to $+0.042$ rubric points per 1,000 characters); (ii) the **length-adjusted intercept** — the expected OE-minus-frontier score gap at *equal length* — is statistically indistinguishable from the raw gap on every axis (accuracy -0.12 [$-0.17, -0.08$] adjusted vs -0.127 raw); and (iii) the accuracy win-difference stays negative **whether OE's answer is longer (-34.3) or shorter (-19.9) than its opponent's** — OE loses on accuracy even when it is the longer answer.

Table 3. Length sensitivity of the rubric cell (n=150; 2,000-replicate cluster bootstrap). Native-scale OE-minus-frontier score gap: raw vs length-adjusted (intercept at equal length); and the accuracy win-difference split by whether OE is the longer or shorter answer.

Axis	Raw gap (pts)	Length-adjusted gap [95% CI]	Slope (pts / +1k chars)	Win-diff: OE longer	Win-diff: OE shorter
Accuracy	-0.127	-0.12 [$-0.17, -0.08$]	-0.015	-34.3	-19.9
Clinical utility	-0.021	-0.03 [$-0.06, +0.01$]	$+0.016$	-14.1	-8.4
Source quality	$+0.103$	$+0.09$ [$+0.03, +0.15$]	$+0.042$	$+15.9$	$+6.0$
Completeness	$+0.004$	-0.01 [$-0.05, +0.03$]	$+0.039$	$+0.7$	-10.8
Verifiability	$+0.003$	-0.01 [$-0.06, +0.05$]	$+0.025$	$+1.4$	$+0.0$

Length has a small positive association with source-quality/completeness/verifiability scores (slopes +0.03 to +0.04 per 1,000 chars) but essentially none with accuracy, and adjusting for it leaves every axis's OE-vs-frontier gap unchanged within CI. The accuracy reversal is therefore **not** explained by length. (This does not make length irrelevant in general — a fully length-matched *generation* study is still pre-registered, §7 — but it removes length as an alternative explanation for the observed reversal.)

5. Critiques of both source studies

The critiques below are **independent of the instrument existence proof** (§4): they hold whether or not the instrument drives the disagreement, and they apply symmetrically to both source studies. We include them because they bound how far *either* published leaderboard can be trusted for procurement, but a reader interested only in the instrument result can treat this section as standalone context. Full detail in `CRITIQUES.md`. Points (a)–(b) concern the systems and settings compared; points (c)–(e) concern the instruments themselves and bear directly on interpretation:

(a) The comparator set omits the AI clinicians actually use. Both studies benchmark **bare frontier API models**; neither includes the **ChatGPT product** (consumer or clinician deployment), whose system prompt, retrieval/browsing, memory, and safety guardrails move the very axes under test. This is especially asymmetric because OE is itself a curated clinical *product*; Nature further queries clinical tools via browser but frontier models via API. A fair benchmark should include a **product arm** distinct from the raw model endpoint.

(b) Reasoning effort is unreported or uncontrolled. Nature reports temperature 0.0 and a seed but **never states reasoning effort** (its cost table confirms reasoning tokens were spent, at an unspecified level); Real-POCQi reports only that "thinking was automatically determined by the LLM." These are two *distinct* undisclosed knobs — whether extended thinking is **on at all**, and if so at what **level (high/medium/low or an explicit token budget)** — and reasoning effort is the largest controllable performance lever for these models, able to swing double digits on exactly the benchmarks in play. Two consequences follow. *Across studies*, neither head-to-head is reproducible on this axis, and their absolute numbers are not comparable. *Within* a study, "automatic" or unstated effort is worse than merely non-reproducible: if a study's thinking setting is auto-determined per query or per model, it may **not be held constant across the very systems it is comparing** — so even that study's internal ranking may reflect an uncontrolled effort difference rather than a capability difference. (Note, too, that temperature 0.0 is largely inert for reasoning models, which sample the reasoning trace regardless; reporting it can create a false impression of determinism.) Our study instead pins effort = high for all four judges and **verifies** token consumption on the real task (§3.2), so the instrument comparison here is not vulnerable to this confound.

(c) The winning margin sits inside the rubric's own noise floor, and the 1–4 scale compresses it. Two linked problems undercut how decisively Nature's rubric separates the systems. *First, a noise-floor problem.* On the real-clinical-question benchmark the frontier systems finish at Gemini 3.62 / GPT 3.54 / Claude 3.52 — a spread of **0.10 of one point across the top three** — with OE at 3.24, i.e. roughly 0.3 point below an essentially indistinguishable frontier cluster, on $n = 100$ questions graded at Krippendorff $\alpha \approx 0.10$ – 0.20 . At that agreement level and sample size the between-frontier ordering is within measurement noise (the "winner" among Gemini/GPT/Claude is not robustly identified), and even the OE gap is a fraction of a rubric point; the study's own low α makes the fine-grained leaderboard order fragile. *Second, a ceiling/compression problem that is also a mechanism for our reversal.* A 1–4 integer rubric has almost no headroom for strong answers — competent clinical responses pile up at 3–4, so the instrument cannot express *how much* better one good answer is than another, and small stylistic preferences (structure, breadth, hedging) get magnified into rank order because the usable dynamic range is one or two points. This compression is exactly why a mean gap of -0.127 point becomes a -29 pp win-difference in our cell C (§3.3a): the rubric floor/ceiling turns near-ties into decisive-looking orderings. A coarse absolute rubric is therefore a **low-resolution instrument for near-parity systems**, and both the Nature leaderboard and our rubric

re-expression inherit that low resolution. This is not a reason to prefer pairwise uncritically — §5(d) levels a symmetric charge at the preference instrument — but it bounds how much weight the Nature rubric ordering can carry.

(d) The pairwise instrument has symmetric, opposite biases — it is not the "true" instrument either. Nothing here should be read as vindicating forced-choice preference as ground truth. Pairwise preference is known to reward surface features — answer length, formatting, citation presence, confident tone — independently of correctness, and Real-POCQI's own data show the mechanism: OE's between-groups accuracy "halo" of +11.4 pp **collapses to +1.5 pp (NS)** once we condition on questions rated both with and without citations (§4.4), i.e. much of the measured preference edge tracks the *presence of citations* rather than answer accuracy. A preference instrument that can be moved 10 pp by attaching references is as construct-contaminated, in the opposite direction, as a compressed rubric. The honest position is that **neither instrument is a clean measure of clinical quality**: the rubric over-rewards frontier-style breadth and compresses near-ties; pairwise over-rewards citations and verbosity. Our claim is the *relative* one — that switching between these two biased instruments is sufficient to flip the winner — not that either endpoint is correct.

(e) The five rubric axes are not independent, so "reversed on k axes" overstates the evidence. Source quality and verifiability both essentially ask "*are there citations, and are they real,*" and accuracy, completeness, and clinical utility are strongly co-scored; the axes are near-collinear rather than five independent measurements. Two consequences: per-axis "significance" counts should not be read as five independent confirmations (a single latent "cited, thorough, frontier-style" factor drives much of the axis structure in both instruments), and this non-independence is *why* we lean on the accuracy axis and the decomposition rather than tallying axes. It also tempers Real-POCQI's own multi-axis sweep: OE winning "on all five axes" is closer to winning on **one or two underlying constructs** measured five ways. We treat the axis set as a small number of correlated constructs throughout, not as five degrees of freedom.

(f) Both studies' data pipelines are run by an interested party, with symmetric — not one-sided — confounds. We have stressed the provenance symmetry (each sourced queries from the platform that won); three further data-pipeline facts deserve naming because they cut in *different* directions and a fair reconciliation must not weaponize only the ones that suit its thesis. (i) *End-to-end control by the winner.* Real-POCQI's pipeline was run by OpenEvidence: OE selected the query pool ($\approx 3,600 \rightarrow 620$ after filtering) from its **own** platform traffic, paraphrased queries via an LLM (Opus 4.6) for de-identification, and paid the physician raters — a chain in which the party under study controls sampling, wording, and rater incentives end to end. Nature's RCQ pipeline is analogously insider-run (NYU Langone's own GPT deployment). Neither is neutral. (ii) *A rater-familiarity confound that actually supports the instrument thesis.* Roughly **half of Real-POCQI's raters were themselves OpenEvidence users**, and the reported OE accuracy edge is **larger among OE-users ($\approx +33.8$ pp) than non-users ($\approx +22.6$ pp)** — a habituation/familiarity gradient that inflates the human-pairwise cell (cell A) for reasons unrelated to answer correctness. This *strengthens* our reading: part of cell A's OE edge is rater familiarity with OE's house format, exactly the kind of preference-instrument artifact that an absolute rubric does not reward. (iii) *Very low response and completion rates.* Real-POCQI's physician recruitment yielded low single-digit response and completion fractions (on the order of $\sim 1\text{--}2\%$), so the rating panel is a self-selected slice of clinicians; combined with (ii), cell A's preference signal carries a selection/familiarity component we cannot remove from public data. None of this shows OE's answers are worse — it shows the *human-pairwise* cell is itself confounded, which is why we do not treat cell A as ground truth and lean instead on the *within-panel* instrument decomposition (§4.5).

(g) On the fine-grained order, both leaderboards agree more than they disagree. Stripped to what is *robust*, both studies essentially resolve only the extremes: OE is top under pairwise and GPT-class models are strong under rubric, but the middle of each ranking is within noise (§5c). Real-POCQI's frontier trio is near-parity behind OE; Nature's frontier trio (Gemini 3.62 / GPT 3.54 / Claude 3.52) is near-parity ahead of OE. The genuine, reproducible cross-study disagreement is therefore narrow — **where OE sits relative to a tightly-bunched frontier cluster** — and it is precisely that one placement that the

instrument flips. This bounds the stakes: we are not reconciling two wildly different capability orderings, but one contested boundary between OE and a frontier pack that both studies otherwise order similarly.

A corroborating confound: Real-POCQi tested **newer** frontier weights (GPT-5.5, Opus 4.8) than Nature (GPT-5.2, Opus 4.6) and still found them losing on human pairwise — a pattern more consistent with an instrument effect than a model-version effect.

6. Discussion

Holding queries and answers fixed, the instrument alone is sufficient to reverse the central claim of a published clinical-AI benchmark. The 2×2 decomposition (§4.5) makes this attribution explicit rather than inferential: because the *same* LLM panel that reverses OE under the rubric **reproduces the human pairwise verdict when it uses the pairwise instrument** (OE winning on four of five axes), the swing localizes to the pairwise→rubric change (C–B, –30.5 to –44.4 pp) and not to the human→LLM change (B–A).

Two scoping caveats bound this claim. *First*, the decomposition is established **within LLM raters**: we have cells A = {pairwise, human}, B = {pairwise, LLM}, and C = {rubric, LLM}, but **not** the fourth cell D = {rubric, human}. The two-study reconciliation is ultimately a claim about a *human-rated* rubric study (Nature), so generalizing our within-LLM instrument effect (C–B) to explain Nature requires assuming **no rater×instrument interaction** — that physicians administering a rubric would show the same pairwise→rubric shift LLMs do. That assumption is plausible but untested, and this is precisely the direction where interaction is likely: humans may not share the frontier-prose prior that an LLM-scored rubric appears to reward, so the human-rubric cell could differ materially from our LLM-rubric cell. We therefore state the proven claim narrowly ("the instrument dominates the rater *modality* among LLM raters") and treat the bridge to Nature as conditional on additivity; filling cell D — even with a small human-rubric sample — is the single highest-value follow-up. *Second*, the axis where our headline lives, accuracy, is also the axis where the rater term is not negligible in the same direction: instrument –32.7 pp and rater –6.8 pp (≈83% instrument), cleaner than before once cell A is put on the same sample, but not a pure instrument effect. The clean "instrument, not rater" statement is strongest for source quality/verifiability/completeness and merely dominant — not exclusive — for accuracy.

We are not the first to suspect the instrument — Real-POCQi says so itself. Real-POCQi explicitly hypothesizes that its divergence from rubric-style benchmarks is *instrument-driven*, and cites JudgmentBench²⁵ — which reports that in a high-expertise professional (legal) domain **head-to-head preference recovers expert judgment more faithfully than absolute rubric scoring** — to argue that its own pairwise design is the more valid one. We credit this: our contribution is not the *idea* that the instrument matters but an **executed, controlled demonstration** of it (queries and answers held fixed) plus a decomposition separating instrument from rater. But we also engage the directional claim honestly, because it cuts against our symmetric "neither instrument is truth" framing (§5d): if JudgmentBench is right that head-to-head is the higher-fidelity instrument in expert domains, then the pairwise (OE-favoring) result deserves *more* evidential weight than the rubric (frontier-favoring) result, and our finding would be not merely "the instruments disagree" but "the more valid instrument favors OE." We do **not** assert this — JudgmentBench's generalization to point-of-care clinical answers is untested, our rubric is LLM- rather than human-administered, and a compressed 1–4 scale is a weak version of rubric scoring — but we flag it as the key open question the missing human-rubric cell (D) would help settle: if physicians administering a rubric still down-rank OE, the disagreement is instrument-intrinsic; if they do not, it localizes to LLM-administered rubrics specifically. Either way, adjudicating *which* instrument is closer to clinical truth (not merely that they differ) is the natural next study, and JudgmentBench gives a prior that pairwise may win that contest.

This does not show OE is worse (or better) than frontier models in the clinic — we have **no adjudicated ground truth** — but it shows that "which system is best" is a function of the measuring instrument, and that pairwise-preference and absolute-rubric instruments encode materially different value functions (preference

rewards OE's clinician-tuned framing and citations; rubric scoring rewards frontier models' breadth and structure). It also reframes the "LLM-judge self-preference" worry: the pro-frontier tilt is a property of the **rubric instrument**, not of LLM judges per se, since those same judges favor OE under pairwise — including the family-neutral judge. Benchmarks that do not report the instrument as an experimental factor — and that do not control reasoning effort, comparator product-vs-endpoint status, and length — are under-specified for procurement decisions.

7. Pre-registered extension (proposed design, not an executed result)

The executed existence proof identifies the instrument; a full **2×2×2 provenance × instrument × citations** factorial (protocol in `reconciliation_protocol.md`) estimates each factor's causal share, including a randomized citation-halo arm (motivated by §4.4), a length-matched *generation* sub-study (to confirm causally what §4.6 shows observationally — that length does not drive the reversal), a product-vs-endpoint arm (§5a), and reasoning-effort sweeps (§5b). Power is grounded in the measured question-level SD (0.598): ≈ 250 questions/cell for a 15 pp interaction, ≈ 390 /cell for 12 pp. Because the Nature RCQ corpus is not public, the provenance arm uses a constructed surrogate LLM-platform query corpus; Real-POCQi is directly reusable. **This surrogate is itself a threat to validity:** a corpus we assemble to *stand in* for NYU's HIPAA-restricted RCQ distribution may not reproduce its topic mix, difficulty, or phrasing, so the provenance-arm estimate is only as trustworthy as the surrogate's fidelity to the true LLM-platform query stream. We therefore treat the provenance factor as the **weakest-identified** arm of the proposed factorial and would pre-register the surrogate-construction procedure and a sensitivity analysis over plausible corpus definitions rather than reporting a single provenance estimate as if the corpus were the real thing.

8. Limitations

(i) **No ground truth** — we show instruments disagree, not which is correct. (ii) **Length not normalized** (OE 3.8k vs GPT 2.6k chars) — we show *observationally* (§4.6) that the reversal is not a length artifact (OE is the longest provider yet loses; length-adjusted gaps match raw gaps within CI), but a fully length-matched *generation* study is still pre-registered to settle it causally. (iii) **House effects** — contestant-family judges self-prefer (quantified; controlled via leave-one-judge-out), and the panel may share a rubric-specific pro-frontier stylistic prior that leave-one-judge-out cannot remove; the pairwise cell (§4.5) shows this prior is instrument-specific (the same judges favor OE under pairwise), and the remaining control — human-administered rubric scoring — is pre-registered. (iv) **Noisy instrument** — modest inter-judge agreement. (v) **Missing fourth cell and the additivity assumption** — we have three of the four 2×2 cells; the decomposition is therefore identified *within LLM raters*, and bridging the instrument effect to the human-rated Nature rubric assumes **no rater×instrument interaction** (that physicians doing a rubric would show the same pairwise→rubric shift LLMs do). This is untested and is the likely direction of interaction (humans may not share the frontier-prose prior); filling cell D = {rubric, human}, even at small n, is the top follow-up. Relatedly, cell A is Real-POCQi's human pairwise protocol while cell B is our LLM pairwise reimplement, so the rater term B–A absorbs minor **protocol** differences (tie handling, pair structure) as well as rater modality — it is "rater modality + protocol," not pure rater. Both cell A and cell B are computed on the identical 150-query sample, so B–A is at least free of sample-composition confounding. The n=150 existence-proof scale is decisive for the LLM-rubric verdict (cell C significantly negative on accuracy and clinical utility) but underpowers the same-sample human baseline (three axes' human CIs cross zero in the subsample, though the full-data values are significant and same-signed); the null-collapsing axes await the full run. (vi) **No rater-level modeling** — the public ratings table has no rater identifier, so human-side rater random effects cannot be fit from public data. (vii) **LLM judges are not bit-deterministic**; we quantify rather than assume stability.

9. Ethics, conflicts, funding

No human subjects or PHI were used; all human ratings are de-identified public data (CC BY 4.0). No Nature-copyrighted data were redistributed (CC BY-NC-ND). **Conflicts of interest:** The author is the founder of Kinnectum AB, an independent medical-AI research venture. The author declares no financial interest in, and no commercial or consulting relationship with, any of the evaluated systems or their providers (OpenEvidence, OpenAI, Anthropic, xAI, Google, or Wolters Kluwer/UpToDate). For transparency we note the source studies' own disclosed conflicts, symmetrically: Real-POCQI's data collection was run by OpenEvidence (the platform under study, which also won it), and the Nature Medicine senior author reports disclosed industry equity/consulting including **Google** — whose Gemini model is the top scorer on that study's headline RCQ leaderboard. Both disclosures are proper and neither implies misconduct; we flag them only because in each study the disclosed interest overlaps the declared winner, which is exactly the kind of provenance/COI symmetry this reconciliation is built to treat even-handedly. **Funding:** This work was funded by Kinnectum AB, which covered the LLM-judge API costs; the funder had no role in study design, analysis, or the decision to publish. **Author contributions:** K.A. designed the study, wrote the analysis and judging code, performed all analyses, and wrote the manuscript (sole author).

10. Data and code availability

All code and outputs are in this repository; `run_all.sh` reproduces every result end to end (`SKIP_JUDGES=1` for the no-API-key public-data subset). Real-POCQI data: Hugging Face `jjfenglab/Real-POCQI` (CC BY 4.0), fetched by `fetch_data.py` and vendored in `data/`. Judge configuration, reasoning-token verification, grades, bootstrap CIs, and figures are regenerable as documented in `README.md`. A machine-readable **per-item instrument-disagreement export** (`judge/export_disagreement.py` → `judge/out/instrument_disagreement.csv`) gives, for every (question, axis, comparator) triple, the rubric and pairwise winners/margins, the number of contributing judges, judge dispersion, and an `instrument_flip` flag, so readers can inspect *where* reversals concentrate; re-aggregating it reproduces the sign of every reported win-difference (a deterministic unit test pins the flip definition). This is a descriptive audit layer, not adjudicated clinical truth. Nature Medicine numbers are cited from the published article (s41591-026-04431-5); its RCQ corpus is not public (IRB i23-00510).

References

Citations appear as numbered footnotes at the point of use and are collected in full below. Numbering follows order of first appearance (Vancouver style); author lists of six or more are abbreviated with *et al.* per Vancouver convention.

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